

TrES-1b

R-1690

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_160819 · created 2026-08-02T13:08:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457619.7994 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.049
Transit depth (Rp/Rs)^2	1.4e-05 +/- 0.0037 [%]
Orbital Inclination (inc)	86.62 +/- 1.47
Airmass coefficient 1 (a1)	0.5 +/- 0.29
Airmass coefficient 2 (a2)	0.46 +/- 0.14
Scatter in the residuals of the lightcurve fit is	54.41 %
Best Comparison Star	None
Optimal Aperture	14.13
Optimal Annulus	20.65
Transit Duration (day)	0.07 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.049 vs 0.1358 (deviation 1.17σ)
Duration fit vs published	0.07 d vs 0.1042 d (deviation 0.6σ)
Mid-time O–C	-0.1 min
Depth SNR	0.0
Residual scatter	54.41 %

Light curve

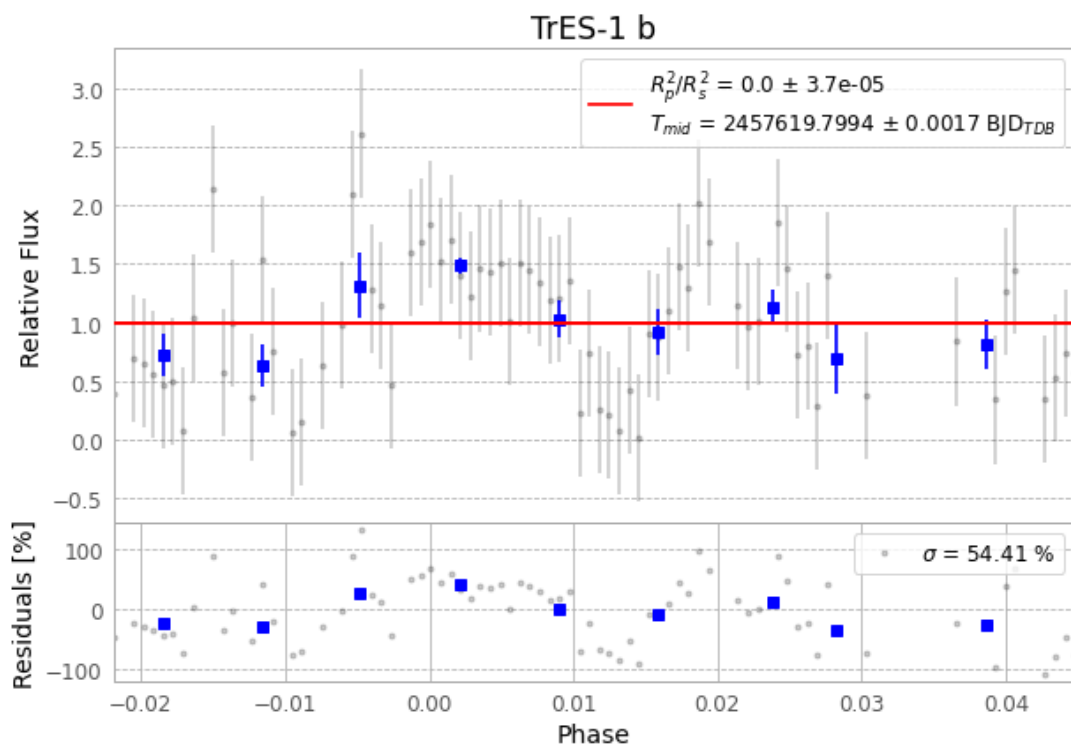


Figure 1 — FinalLightCurve_TrES-1 b_2016-08-18.png (SHA-256 7d9cf20f25e48de4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 436], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aaa31e4022a7569f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

100 FITS frames (66 MB), TRES-1160819053012.FITS → TRES-1160819102712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-TRES-1-160819.log (7af1dcd9b2f4...), FinalLightCurve_TrES-1 b_2016-08-18.png (7d9cf20f25e4...), FinalLightCurve_TrES-1 b_2016-08-18.csv (e3913387f0a7...)

Compute resources

Wall time 435.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 13:08Z.